

1. Fundamental Concepts in Cell Biology

1.1 Cell theory & modern view

- Cell is the smallest unit that can perform all life processes (metabolism, growth, response, reproduction).
- Modern additions:
 - o Energy flow occurs within cells (respiration, photosynthesis).
 - o Hereditary information (DNA → RNA → protein) is housed and expressed in cells.
 - o All cells share basic biochemical machinery (ATP, DNA, RNA, ribosomes, membranes).

1.2 Prokaryotic vs eukaryotic cells (mechanistic view)

Prokaryotes

- Genome: single circular DNA; often supercoiled; no histones (bacteria) or archaeal type histones.
- Compartmentation: no internal membrane systems; transcription and translation are coupled.
- Surface: cell wall (peptidoglycan in bacteria), outer membrane in Gram negative, capsules, flagella (flagellin), pili.
- Energy: plasma membrane houses electron transport chains.

Eukaryotes

- Genome: multiple linear chromosomes packaged with histones into chromatin; transcription and translation decoupled (nucleus vs cytoplasm).
- Compartmentation: nucleus and organelles with specific microenvironments → spatial separation of pathways.
- Cytoskeleton: actin, microtubules, intermediate filaments; basis for shape, division, organelle positioning, motility.
- Endomembrane system: ER, Golgi, lysosomes, endosomes, vesicles.

2. Plasma Membrane & Cell Surface

2.1 Plasma membrane structure

- Fluid mosaic model: phospholipid bilayer with embedded proteins, cholesterol (in animals), and carbohydrates on outer face.
- Lipids: phospholipids (amphipathic), glycolipids, cholesterol.
- Proteins: integral (span bilayer, e.g. transporters, receptors) and peripheral (bound

Prokaryotes

From Greek pro- "before" and karyon "nut, kernel" (referring to the nucleus). Literally means "before-kernel" or "pre-nucleus," signifying cells lacking a true nucleus. The term was established in the 1960s to distinguish them from eukaryotes.

Eukaryotes

From Greek eu- "good, well, true" and karyon "nut, kernel" (referring to the nucleus). Literally means "true-kernel" or "true-nucleus," signifying cells having a membrane-bound nucleus. It describes the more complex cells of animals, plants, fungi, and protists.

Functions

- Selective permeability: controls entry/exit of ions, nutrients, waste.
- Signal transduction: receptors for hormones, growth factors, neurotransmitters.
- Cell-cell recognition and adhesion: glycoproteins, glycolipids, CAMs (cadherins, integrins).
- Scaffold: anchors cytoskeleton and extracellular matrix (via integrins).

2.3 Membrane transport

- Passive transport:
 - o Simple diffusion (gases, small hydrophobic molecules).
 - o Facilitated diffusion (via channels or carriers; no ATP; down gradient).
- Active transport:
 - o Primary (ATP driven; e.g. Na^+/K^+ ATPase).
 - o Secondary (uses existing ion gradients).
- Vesicular transport:
 - o Endocytosis (phagocytosis, pinocytosis, receptor mediated).
 - o Exocytosis (regulated and constitutive secretion).

3. Cytoplasm, Cytoskeleton, Cell Wall, ECM

3.1 Cytoplasm & cytosol

- **Cytosol**: crowded solution of proteins, metabolites, ions; site of glycolysis, many biosynthetic reactions.
- Highly organized: microdomains, phase separation, cytoskeletal tracks.

3.2 Cytoskeleton

Microfilaments (actin)

- Structure: two stranded helical polymers; polar; often anchored to membrane.
- Functions: cell shape, cortical tension, lamellipodia/filopodia, muscle contraction (with myosin), cytokinesis (contractile ring).

Microtubules

- Structure: hollow tubes of α/β tubulin dimers; plus and minus ends; dynamic instability.
- Functions:
 - o Mitotic spindle; **chromosome** segregation.
 - o Organelle positioning and vesicle transport (kinesins toward + end, dyneins toward

Cytosol

From Greek *kytos* "a hollow vessel" (cell) and Latin *solvere* "to loosen or dissolve" (via "-sol"). It is the liquid part of the cytoplasm, essentially the "cell-solution" or "cell-dissolved" substance. The term was coined to differentiate the liquid from the organelles suspended within it.

chromosome

From Greek *chroma* "color" and *soma* "body." Literally translates to "color-body," due to their strong affinity for staining dyes used in microscopy. The term was introduced in the late 19th century by Heinrich von Waldeyer-Hartz.

lamins).

- Mechanical strength; nuclear lamina; **desmosomes** and hemidesmosomes.

3.3 Cell wall (plants, fungi, bacteria)

Plants

- Composition: cellulose microfibrils in matrix of hemicellulose, pectin, proteins; often lignin (secondary wall).
- Functions: tensile strength, protection, turgor maintenance, cell shape, defense.

Fungi: chitin, glucans, mannoproteins.

Bacteria: peptidoglycan; Gram positive vs Gram negative architecture.

3.4 Extracellular matrix (ECM) in animals

- Components: fibrous proteins (collagen, elastin), glycoproteins (fibronectin, laminin), proteoglycans.
- Functions: mechanical support, substrate for cell migration, reservoir of growth factors, signaling via integrins (outside in and inside out).

4. Nucleus and Genetic Organization

4.1 Nuclear envelope & nuclear pores

- Double membrane: outer continuous with rough ER; perinuclear space.
- Nuclear pore complexes (NPCs): large macromolecular assemblies; regulate nucleocytoplasmic transport via signal sequences (NLS, NES) and transport receptors (importins/exportins, Ran GTP).

desmosomes

From Greek desmos "bond, tie" and soma "body." Literally means "bonding-body," referring to their function as spot welds or anchors between cells. They are specialized structures that provide strong adhesion and mechanical stability to tissues.

4.2 Chromatin organization

- DNA wound around histone octamers → nucleosomes (beads on a string).
- Higher order folding → euchromatin (less condensed, transcriptionally active) vs heterochromatin (condensed, inactive or structural).
- Epigenetic regulation: histone modifications, DNA methylation, chromatin remodeling complexes.

4.3 Nucleolus

- Not membrane bound; ribosomal RNA synthesis (by RNA polymerase I) and assembly of ribosomal subunits.
- Has fibrillar centers, dense fibrillar component, granular component.

4.4 Nuclear functions

- Storage and expression of genetic information; regulation of transcription.
- Coordination of DNA replication, repair, recombination.
- Cell cycle regulation via checkpoint machinery localized to nuclear structures.

5. Endomembrane System

5.1 Endoplasmic reticulum (ER)

Rough ER (RER)

- **Ribosomes** bound via receptors; site of synthesis of:
 - o Secreted proteins, membrane proteins, and proteins of ER, Golgi, lysosomes.
- Co translational translocation via signal peptide, SRP, translocon.
- Folding and quality control; N linked glycosylation; disulfide bond formation (protein disulfide isomerase).
- ER stress and unfolded protein response (UPR) when folding capacity is overwhelmed.

Smooth ER (SER)

- Lipid and steroid synthesis, membrane biogenesis.
- Detoxification (cytochrome P450 enzymes) in liver.
- Calcium storage and release (sarcoplasmic reticulum in muscle).

5.2 Ribosomes

- Ribonucleoprotein complexes (large + small subunits).
- Cytosolic (free) ribosomes: proteins for cytosol, nucleus, mitochondria, peroxisomes.

Chromatin

From Greek chroma "color" (referring to staining) and the suffix -in "substance." Literally translates to "color-substance," the material that makes up chromosomes, due to its staining properties. It is the complex of DNA and proteins (histones) found inside the eukaryotic nucleus.

Ribosomes

A blend of ribose (a sugar component of RNA) and Greek soma "body." It is the "RNA-body," a molecular machine essential for protein synthesis (translation). The term highlights their composition of ribosomal RNA (rRNA) and protein.

- Membrane bound (on RER): proteins for secretion, membranes, endomembrane organelles.

5.3 Golgi apparatus

- Flattened cisternae: cis Golgi (toward ER), medial, trans Golgi network.
- Functions:
 - o Modifies N linked oligosaccharides, adds O linked glycosylation.
 - o Sorts proteins based on signal motifs into different vesicles (secretory, lysosomal, plasma membrane).
 - o Produces lysosomal enzymes tagged with mannose 6 phosphate.
- Key for polarized trafficking and cell surface composition.

5.4 Lysosomes

- Acidic organelles (pH ~5) with hydrolytic enzymes (proteases, nucleases, glycosidases, lipases, phosphatases).
- Pathways:
 - o Endocytosis → early endosome → late endosome → lysosome.
 - o Autophagy: macroautophagy (double membrane autophagosome), microautophagy, chaperone mediated autophagy.
- Roles: degradation and recycling of macromolecules, organelle turnover, response to starvation.
- Lysosomal storage diseases: defects in specific enzymes → accumulation of substrates.

5.5 Endosomes

- Early endosomes: sorting station; decide between recycling to membrane vs degradation.
- Late endosomes/multivesicular bodies: intermediates before lysosomes.
- Crucial in receptor downregulation, signaling termination or propagation.

6. Peroxisomes, Vacuoles, Storage Organelles

6.1 Peroxisomes (microbodies)

- Single membrane organelles; contain oxidases and catalase.
- Functions:
 - o β oxidation of very long chain fatty acids (in animals).

Lysosomes

From Greek lysis "to loosen, dissolve" and soma "body." Literally means "dissolving-body," referring to their role in breaking down waste materials and cellular debris. They contain hydrolytic enzymes and function as the cell's waste disposal system.

Peroxisomes

A blend of peroxide (referring to hydrogen peroxide) and Greek soma "body." Literally the "peroxide-body," as they are the site of reactions that generate and then safely break down hydrogen peroxide. They are tiny organelles involved in various metabolic processes, including fatty acid breakdown.

seeds.

- Assembly: import of matrix proteins with PTS signals from cytosol; growth and division.

6.2 Vacuoles (mostly in plants, fungi)

- Large central vacuole in mature plant cells; can occupy most of cell volume.
- Functions:
 - o Storage (ions, metabolites, pigments, defensive compounds).
 - o Osmoregulation and turgor maintenance.
 - o Degradative functions (lysosome like).
- Tonoplast: vacuolar membrane with specific transporters.

6.3 Storage granules & inclusions

- Glycogen granules, lipid droplets, pigment granules.
- Not membrane bound (e.g. glycogen) or monolayer bound (lipid droplets).
- Energy reserves, thermal insulation, signaling platform (lipid droplets).

7. Mitochondria & Chloroplasts (Endosymbiotic Organelles)

7.1 Mitochondria

- Double membrane: outer smooth; inner folded into cristae; intermembrane space; matrix.
- Own circular DNA, ribosomes (bacterial like), independent replication (binary fission).
- Functions:
 - o Oxidative phosphorylation (ETC in inner membrane; ATP synthase).
 - o Citric acid cycle (matrix), fatty acid β oxidation, amino acid metabolism.
 - o Apoptosis regulation: cytochrome c release; intrinsic pathway.
 - o Ca^{2+} buffering; ROS generation and detoxification.

7.2 Chloroplasts (and other plastids in plants)

- Double membrane; internal thylakoid membranes stacked into grana; stroma.
- Chlorophyll and other pigments in thylakoid membranes.
- Light reactions: electron transport, photophosphorylation, NADPH formation.
- Dark reactions (Calvin cycle) in stroma: CO_2 fixation $\rightarrow$ sugars.
- Plastids differentiation: chloroplasts, chromoplasts (pigments),

Mitochondria

From Greek mitos "thread" and chondrion "small grain" or "granule." Literally means "thread-granules," a name derived from their thread-like or granular appearance under the microscope. They are the "powerhouses" of the cell, generating most of the cell's supply of ATP.

Chloroplasts

From Greek chloros "pale green" and plastos "molded, formed." Literally means "green-forms," aptly describing the green-pigmented, formative organelle in plant cells and algae. They are the site of photosynthesis, responsible for converting light energy into chemical energy.

amyloplasts (starch storage).

7.3 Endosymbiotic theory

- Mitochondria and chloroplasts arose from engulfed bacteria (α proteobacteria and cyanobacteria).
- Evidence: double membranes, circular DNA, bacterial type ribosomes, sensitivity to some antibiotics, division by fission.

8. Centrosomes, Cilia, Flagella, Microvilli

8.1 Centrosomes & centrioles

- Centrosome: main microtubule organizing center (MTOC) in animal cells; near nucleus.
- Contains a pair of centrioles (barrel shaped microtubule triplets) + pericentriolar material.
- Functions: nucleate and anchor microtubules; organize mitotic spindle.

8.2 Cilia and flagella (eukaryotic)

- Axoneme: 9 doublet microtubules + 2 central microtubules ("9+2").
- Powered by dynein motor proteins; bending via sliding of microtubule doublets.
- Primary (non motile) cilia: sensory roles; 9+0 structure.
- Functions: locomotion (single celled eukaryotes), fluid movement across epithelia (respiratory tract, oviduct), mechanosensation.

8.3 Microvilli & other projections

- Microvilli: actin based projections increasing surface area (intestinal epithelium, kidney tubules).
- Stereocilia: long microvillus like structures (epididymis, inner ear hair cells).
- Filopodia/lamellipodia: dynamic actin structures for migration and environmental sensing.

9. Cell Junctions and Communication (brief)

- Tight junctions (occluding): seal cells; maintain polarity and barrier (epithelia).
- Adherens junctions & desmosomes (anchoring): link cytoskeletons of neighboring cells (actin or intermediate filaments).
- Hemidesmosomes: anchor cells to ECM.
- Gap junctions: connexon channels; allow ion/small molecule exchange; electrical and metabolic coupling.
- Plasmodesmata (plants): cytoplasmic channels through cell walls.

10. Cell Cycle, Division & Control

10.1 Cell cycle phases

- G₁: growth, biosynthesis, preparation for DNA replication; decision point (restriction point).

flagella

The plural of flagellum, from Latin flagellum, meaning "whip." The name directly refers to their structure and function as whip-like appendages used for cell motility. They are thin, tail-like structures that allow cells (like sperm or bacteria) to swim.

- S: DNA replication; histone synthesis; centrosome duplication.
- G₂: further growth, preparation for mitosis (check DNA integrity).
- M phase: mitosis (nuclear division) + cytokinesis (cytoplasm division).
- G₀: quiescent state; reversible (e.g. liver) or terminally differentiated (neurons, muscle).

10.2 Mitosis (mechanistic view)

- Prophase: chromatin condensation; centrosomes move apart; spindle begins to form; nuclear envelope intact.
- Prometaphase: nuclear envelope breakdown; spindle microtubules attach to kinetochores.
- Metaphase: chromosomes aligned at metaphase plate; bipolar attachment under tension.
- Anaphase: cohesion cleaved; sister chromatids segregate to opposite poles; spindle elongates.
- Telophase: chromosomes decondense; nuclear envelopes reassemble; contractile ring constricts.
- Cytokinesis: actomyosin ring divides cell into two.

10.3 Meiosis (very short, for contrast)

- Two successive divisions after one S phase → 4 haploid cells.
- Meiosis I: homologous chromosomes pair and segregate (reductional).
- Meiosis II: sister chromatids segregate (equational).
- Crossing over (prophase I) and independent assortment generate genetic diversity.

10.4 Molecular control: cyclins, CDKs, checkpoints

- CDKs: serine/threonine kinases; activated by binding specific cyclins and by phosphorylation state.
 - Cyclin levels oscillate; CDK levels constant.
 - Major checkpoints:
 - o G₁/S: DNA damage? size? environment (growth factors, nutrients)?
 - o G₂/M: DNA fully replicated undamaged?
 - o Spindle assembly checkpoint: all kinetochores attached and under proper tension?
- Key regulators:

Mitosis

From Greek mitos "thread" and the suffix -osis "process." Literally means "thread-process," referring to the characteristic thread-like appearance of condensed chromosomes during cell division. It is the process of cell division resulting in two genetically identical diploid daughter cells.

Meiosis

From Greek meioun "to make smaller" or "lessen" and the suffix -osis "process." Literally means "lessening-process," referring to the reduction in the number of chromosomes by half. It is the process of cell division that produces haploid gametes (sperm and egg) for sexual reproduction.

repair, or apoptosis.

- Rb: controls G₁/S transition by binding E2F; phosphorylation by cyclin-CDKs releases E2F → S phase gene expression.
- Oncogenes (e.g. Ras, Myc) and tumor suppressor genes (p53, Rb) integrate growth signals and stress.

11. Apoptosis & Cell Turnover (bonus but high yield at graduate level)

- Programmed cell death; characterized by cell shrinkage, chromatin condensation, membrane blebbing, apoptotic bodies; no inflammation.
- Pathways:
 - o Intrinsic (mitochondrial): Bcl 2 family, cytochrome c, caspase 9.
 - o Extrinsic (death receptor): Fas, TNF receptor, caspase 8.
- Maintains tissue homeostasis; removes damaged or potentially oncogenic cells.